

# Temporal Experience and Judgment: Why 'How Long?' Can Be Several Different Questions

*Human-readable companion to Temporal Experience and Judgment: Process History, Represented Extent, and Query-Specific Readouts*

*Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.*

## Abstract

A single interval can support several apparently contradictory temporal answers. While an event is unfolding it may feel slow; later it may seem brief; a story considered in a moment can represent years; a person can misestimate duration yet still read a clock correctly and act on schedule. Those outputs are not necessarily competing measurements of one hidden quantity. They can refer to different objects and use different information. This companion presents the manuscript's dissociation architecture in ordinary language. It separates the history a process actually underwent, the temporal extent represented by current content, the information available to a particular temporal query, the readout produced by that query, and public clock information. It then adds a prospective admissibility rule: a relation may be treated as direct only when the query and the information have the same referent; cross-referent claims require a predeclared transformation with specified inputs, target, basis, and discriminating consequence. Test A asks whether a preregistered represented-extent reconstruction pattern is supported and, separately, whether constraining the model to that pattern improves held-out prediction. Test B asks whether temporal measurement meaning and interface sufficiency survive a change in clock environment. The framework does not claim a new internal clock or deny physical time. It earns value only if its prospective restrictions transfer across tasks and improve inference or prediction beyond established timing and event-memory accounts.

## 1. The human problem: one interval can have several honest temporal answers

Think about a long meeting, a fast-moving conversation, a story that spans decades, and an appointment at 3:00 p.m. Each can involve time, but the question 'how long?' does not always ask for the same thing. A meeting may feel slow while it is happening and then occupy little space in memory. A story can represent twenty years even though reading the sentence took seconds. A person can be poor at estimating elapsed minutes and still use a clock accurately enough to arrive on time.

The originating intuition behind this branch was 'time dilation - internal versus external clock.' The completed paper turns that intuition into a narrower scientific problem. It does not propose that physical time changes for the person or that subjective time is a second fundamental physics. It asks which temporal object a judgment is about, which information was available to the query, and what transformation - if any - connects that information to the reported answer.

This matters because temporal research often uses words such as duration, passage, remembered time, represented time, clock time, and scheduled action near one another. Established work already shows that prospective and retrospective duration judgments can rely on different processes, that event structure and memory influence later duration reports, that semantic content can carry temporal extent, and that clock monitoring can support prospective action. The manuscript does not claim those component dissociations as new. Its residual contribution is a prospective method for preventing one temporal relation from being silently substituted for another.

## 2. What actually happened is not the same thing as what is represented now

The first distinction is between process history and represented temporal content. Process history is the trajectory the modeled system actually went through during an interval. It can contain changes in sensory input, bodily state, task state, memory activation, or other declared variables. Laboratory time indexes that trajectory, but the participant does not have to store or inspect the trajectory as a literal record.

Represented temporal content is different. At a given moment a person can represent an event as lasting minutes, days, or decades. A present thought about a childhood summer can represent months without the current thought itself lasting months. A sentence such as 'ten years later' carries a large represented extent while being processed quickly. The manuscript therefore blocks an inference that the duration represented by content reveals how long the current cognitive process took.

The reverse inference also fails. A process can last a long time without representing that long extent as content. Long processing history and large represented extent are different types of object. Their relationship has to be measured rather than assumed.

### 3. A temporal answer belongs to a query

At the moment a question is asked, the system has some encoded history and some query instruction. The manuscript calls the information usable by that particular temporal computation  $A_q$ . The notation is useful because the same stored material need not be used identically by every question.

A question about felt passage, a retrospective estimate of laboratory duration, a judgment of how much time a narrative represented, a clock-reading task, a calibration judgment, and a scheduled action can all draw on different subsets or transformations of available information. The observed answer  $y_q$  is therefore query-specific. The framework does not infer that every cause of a temporal report was itself a conscious temporal content.

This is where ordinary language can mislead. Saying 'the person knows how long it was' may compress at least several possibilities: the person remembers event structure and reconstructs a duration; the person has a semantic representation of an event's extent; the person saw a clock; the person learned a mapping between internal cues and seconds; or the person can act correctly at the scheduled time. Those are empirically related functions, but they are not identical by definition.

### 4. The prospective admissibility rule: direct relation or declared transformation

The paper's main methodological move is to make the mapping between information and temporal answer explicit before the confirmatory result is known. A relation is referent-direct when the information and the query concern the same temporal object. When they concern different objects, the route is cross-referent and requires a specified transformation.

A cross-referent transformation is not licensed merely by giving it a name such as reconstruction, calibration, compression, or inference. The analyst must declare what information enters the transformation, what temporal target it is supposed to produce, what operational or theoretical basis constrains it, and what empirical consequence would distinguish that restricted transformation from a more flexible alternative.

The purpose is not to outlaw reconstruction. Reconstruction is a mature idea in timing and event-memory research. The purpose is to prevent a later duration report from being treated as a transparent readout of the online process simply because a plausible reconstructive story can be invented after the fact.

### 5. Three levels of claim strength

| Level                              | What it does                                                                                                         | What success would mean                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Level 1: conceptual object typing  | Separates process history, represented content, query information, readout, and clock/interface objects.             | Clarity about what is being measured; not an empirical discovery by itself. |
| Level 2: prospective admissibility | Specifies information lineage, query referent, allowed transformation, and boundary lineage before outcome analysis. | A reusable method for generating constrained hypotheses.                    |
| Level 3: registered implementation | Turns the Level 2 restrictions into explicit models, priors, inequalities, or comparisons for a particular task.     | A falsifiable empirical test of that implementation.                        |

Keeping these levels separate prevents two opposite errors. A useful conceptual distinction should not be advertised as though an experiment discovered the distinction itself. Conversely, a failed implementation should not

automatically erase the conceptual need to distinguish the objects if the failure concerned only one proposed transformation or model restriction.

## **6. Test A: does represented extent help reconstruct a duration judgment in the predicted way?**

Test A carries the main within-boundary empirical burden. It asks whether a preregistered reconstruction pattern linking represented temporal extent to a duration judgment is supported under the chosen task. The manuscript deliberately splits the decision in two.

Decision D1a asks whether the registered positive reconstruction pattern itself is supported. In ordinary language: do the coefficients or effects move in the constrained direction predicted by the represented-extent account?

Decision D1b asks a different question: does forcing the model to obey that pattern improve held-out prediction compared with models that use the same information but are less constrained or differently constrained? A theoretically attractive pattern can be present without buying predictive advantage, and predictive advantage can be tested without pretending that the underlying object distinction was newly discovered.

This separation matters because a framework can otherwise win too easily. If any flexible mapping from remembered event structure to a duration report counts as success, the model can accommodate almost any outcome. Requiring the transformation to be restricted prospectively and then testing whether the restriction helps prediction turns the interpretation into something that can lose.

## **7. Test B: clocks are interfaces, and interface changes need their own test**

The paper treats clock-linked temporal functions separately from reconstructed duration. A clock is an external measurement interface. Seeing 2:55 and later 3:05 can support a ten-minute inference without requiring the same internal information that supported felt passage or retrospective reconstruction.

Test B/D2 asks whether measurement meaning is preserved when the clock environment or interface changes, and whether the information exposed by that interface is sufficient for the intended temporal function. The concern is measurement lineage: if the interface changes its scale, mapping, update rate, or semantics, a successful action cannot simply be compared as though the same temporal information had been presented.

This is especially important for scheduled action. Recognizing a clock value, mapping it to a schedule, monitoring for a target, choosing an action, and experiencing passage are different functions. A clock-speed manipulation that changes prospective-memory performance does not by itself establish what happened to felt time. Conversely, altered felt passage does not imply failure of public-time action.

## **8. Why event-memory and timing theories remain strong competitors**

The architecture is deliberately designed to lose against established accounts when they already explain the result. Event segmentation, contextual change, perceptual-state accumulation, compressed recollection, semantic duration, episodic simulation, prospective-memory clock monitoring, state-dependent timing, and bodily timing all provide mature alternatives in their own domains.

For example, neural and behavioral work shows that event structure and changes during encoding can predict later duration estimates. Other work shows that represented event duration can be carried by language and memory, and that learned event structure can influence prospective judgments. These precedents mean the paper cannot claim that reconstruction, segmentation, or state change are proprietary mechanisms.

A conventional model can also adopt the manuscript's declarations and become empirically equivalent for a particular implementation. That is allowed. The residual value is not a special mechanism but a prospective constraint on which temporal mappings are licensed and how they should transfer across tasks.

## 9. The ordinary observable is insufficient because several temporal functions can agree or disagree

A duration number by itself does not reveal how it was produced. Two people can both answer 'five minutes' while one relied on a clock and the other reconstructed from event count. The same person can answer differently when asked about felt passage, elapsed laboratory duration, or represented narrative extent without any contradiction if the questions have different referents.

Likewise, correct scheduled action is not proof of accurate unaided duration estimation, and a poor retrospective estimate is not proof that the interval was poorly represented in every temporal sense. The paper therefore asks experiments to preserve the query, information lineage, and interface rather than treating all temporal outputs as manifestations of one scalar subjective-time variable.

## 10. What the framework can and cannot establish

If Test A supports D1a, the evidence supports the preregistered reconstruction pattern for that task. If D1b also succeeds, imposing that pattern adds held-out predictive value relative to the specified alternatives. Neither result proves that represented extent is the only determinant of duration judgment, and neither turns the Level 1 distinctions into empirical discoveries.

If Test B/D2 succeeds, the evidence supports preservation of the declared temporal measurement semantics and the sufficiency of the specified clock/interface record under the tested enlargement. It does not establish a universal clock theory or a general claim about conscious time.

The framework does not claim that subjective time is a separate physical substance, that internal processing ceases to occur in ordinary time, or that every temporal judgment arises from one route. It does not explain subsecond intentional binding or heartbeat-linked interval effects within its current scope. Those phenomena would require new implementations rather than being absorbed by an unrestricted readout function.

## 11. How the proposal must shrink or disappear

The paper makes prospective reusability a criterion rather than an assumption. Level 2 earns scientific value only if its predeclared restrictions can be carried into new tasks and expose predictive or transfer structure that established timing accounts do not capture as well.

- If the same-information unrestricted model predicts just as well as the registered reconstruction restriction, D1b fails even if D1a is supported.
- If event-memory, state-dependent timing, semantic-duration, clock-monitoring, or another mature model fully explains the task without the admissibility architecture changing an inference, use the mature model.
- If a supposed cross-referent relation requires a transformation that cannot be specified prospectively or distinguished empirically, do not treat the later readout as direct evidence about the original referent.
- If clock/interface meaning changes across environments and the inherited semantics cannot be preserved, treat the conditions as different measurement lineages rather than one unchanged temporal measure.
- If the Level 2 rules fail to transfer across tasks, the broader methodological claim should be narrowed to the local implementations that survive.

Local failure therefore has to be typed. A failed coefficient restriction can kill that registered implementation without erasing the underlying distinction between process history and represented temporal content. Whole-framework redundancy requires a stronger result: established accounts must do the same inferential and transfer work without the added discipline.

## 12. Conclusion

The paper begins from an ordinary experience: one interval can feel long, be remembered as short, represent years, and still be navigated correctly with a clock. The scientific lesson is not that one of those answers is the real time while the others are illusions. It is that they can be answers to different temporal questions.

The framework therefore separates what the process actually did from what temporal extent is represented now, which information a specific query can use, and what readout or action follows. Cross-referent claims require a declared transformation rather than a retrospective story. Test A asks whether one such reconstruction pattern is supported and whether constraining the model to it helps prediction. Test B asks whether clock/interface semantics survive boundary change.

If those prospective restrictions transfer and improve prediction or prevent invalid inferences, the architecture earns work of its own. If established timing models can do the same job without it, the framework should collapse into those accounts. The contribution is therefore a discipline for keeping temporal referents and information lineages straight, not a new universal clock.

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